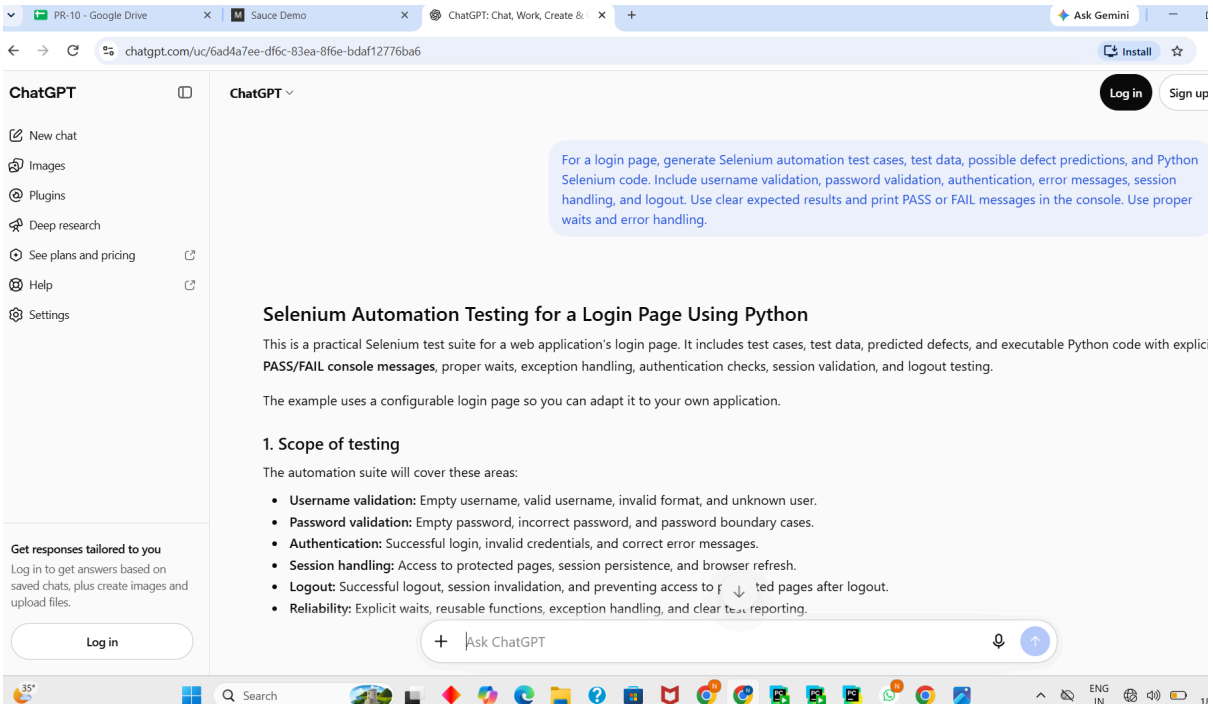
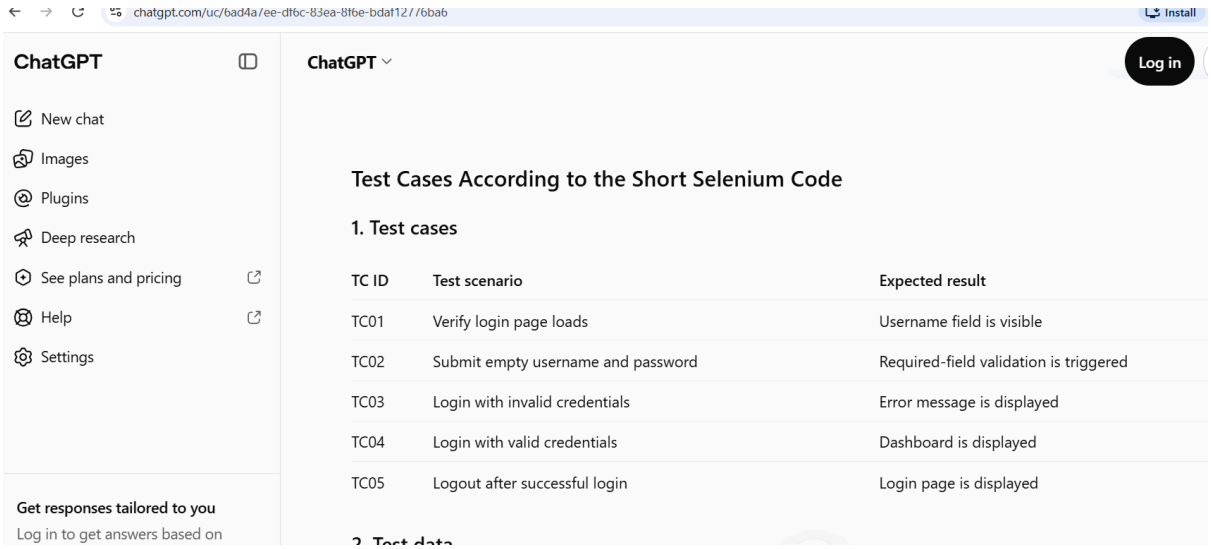
Module Name:AI-Powered Software Testing
Test Case ID:ST_12
Webpage URL:https://sauce-demo.myshopify.com/

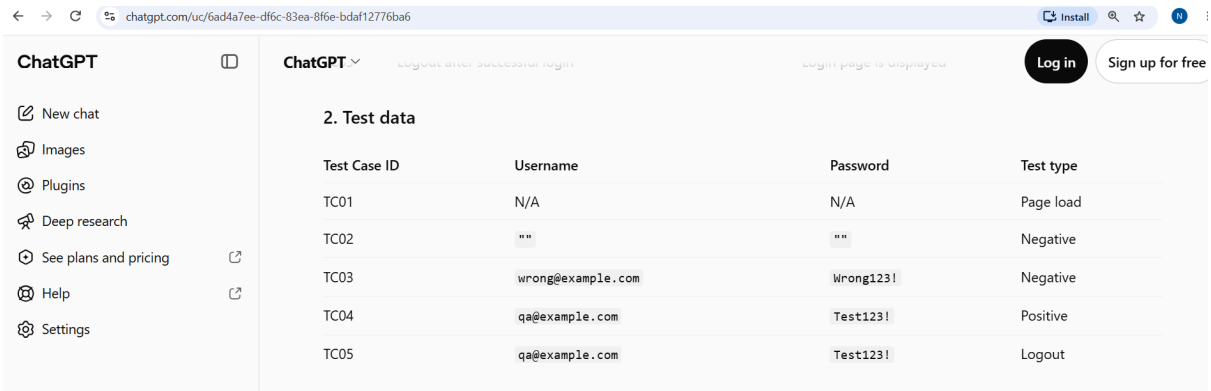
Scenario_ID:TC_01
Scenario:AI Prompt



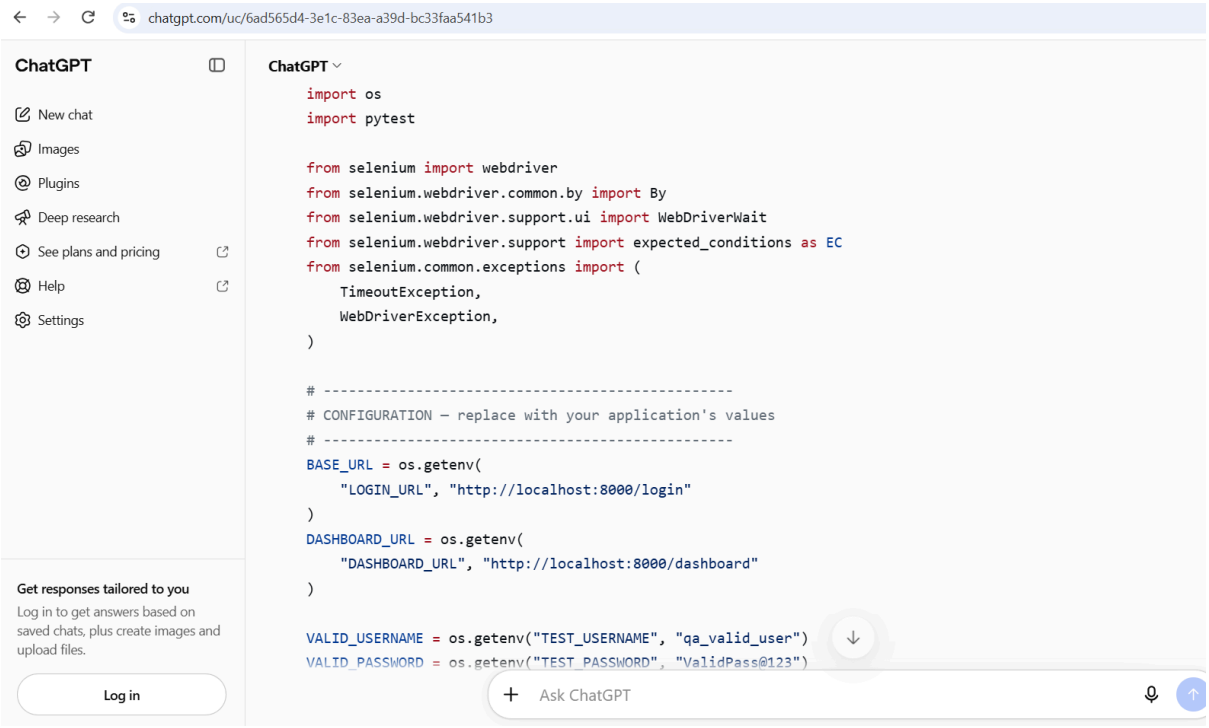
Scenario_ID:TC_02
Scenario:AI-Generated Test Case

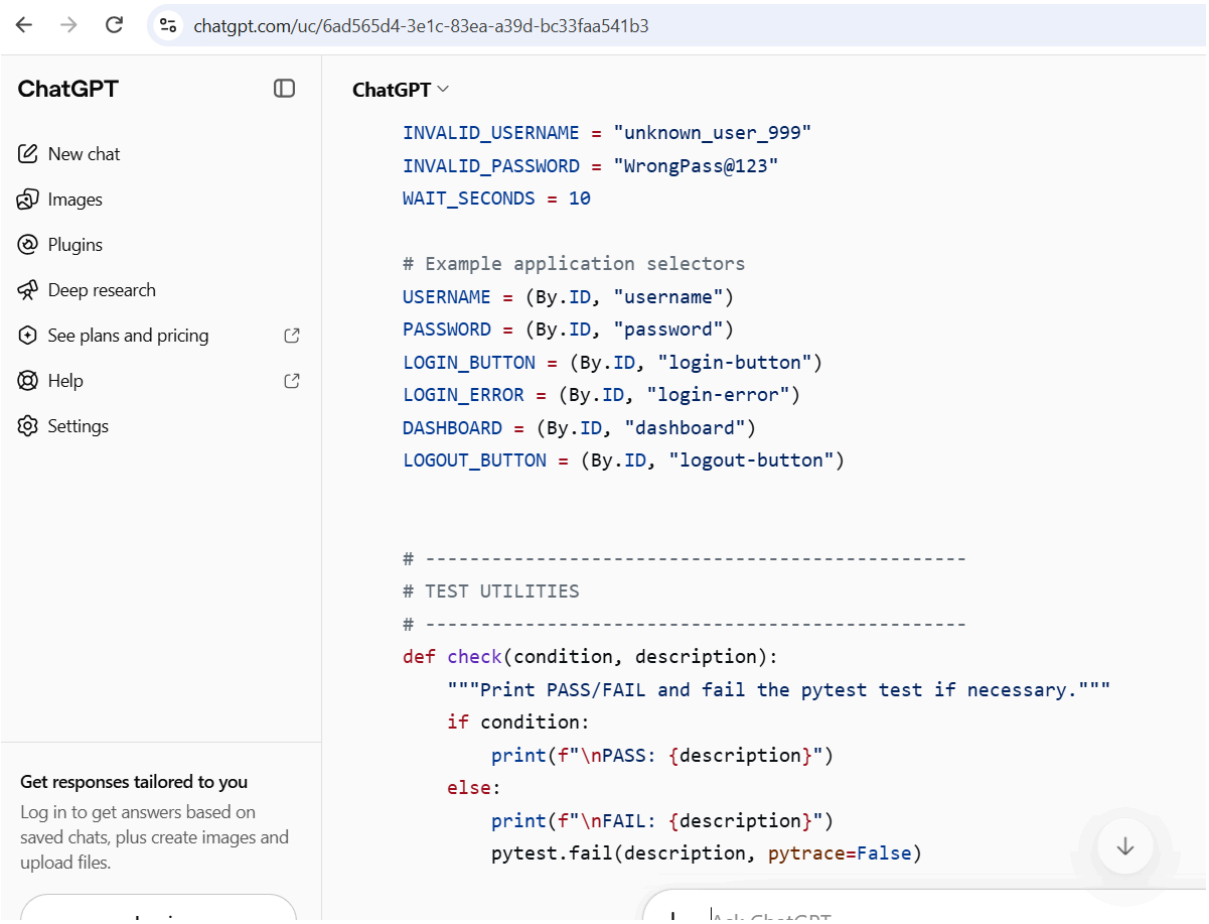


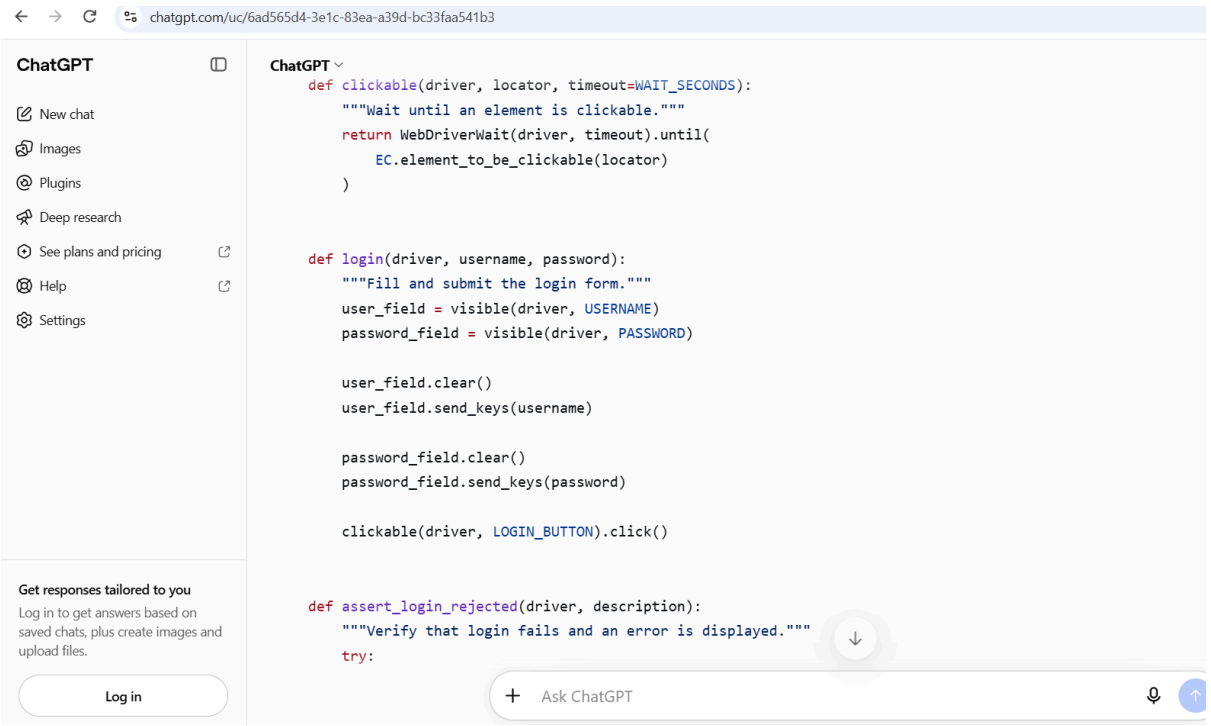
Scenario_ID:TC_03
Scenario:AI-Generated Test Data

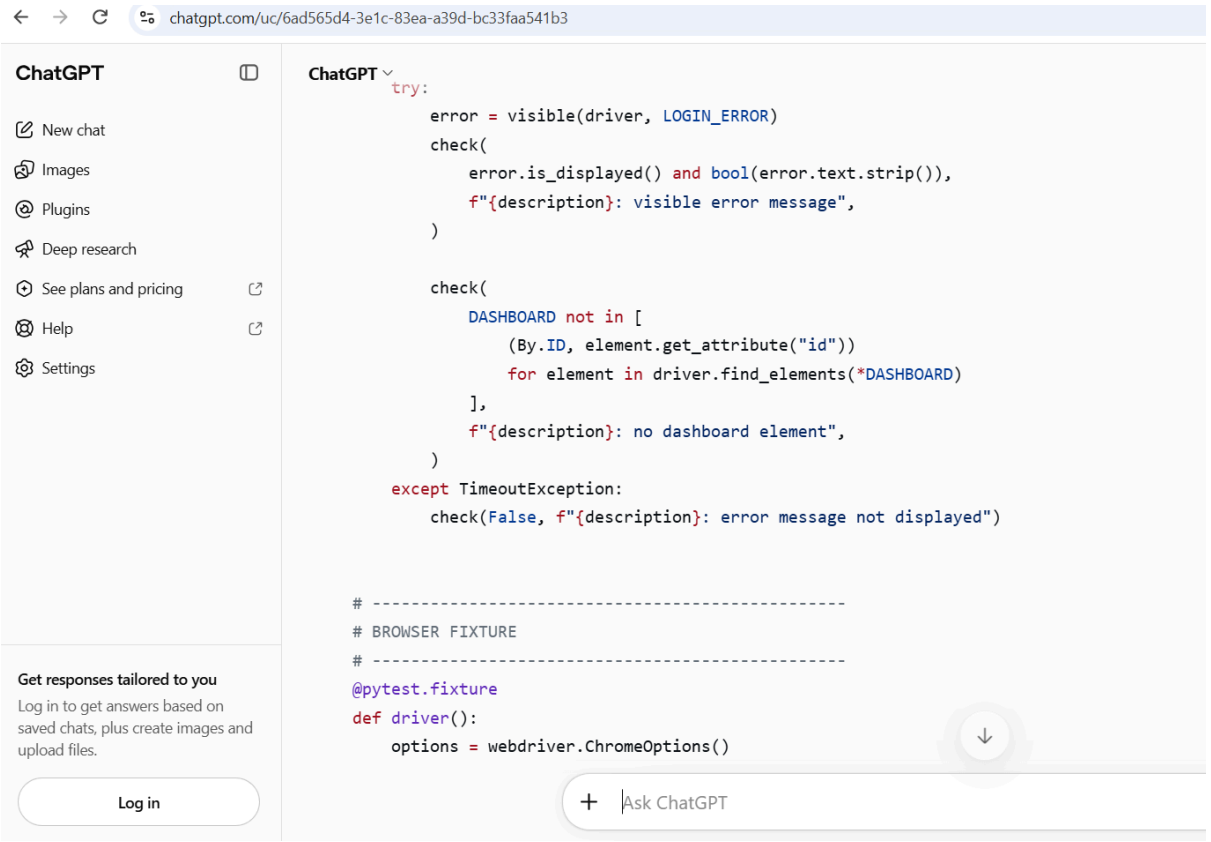


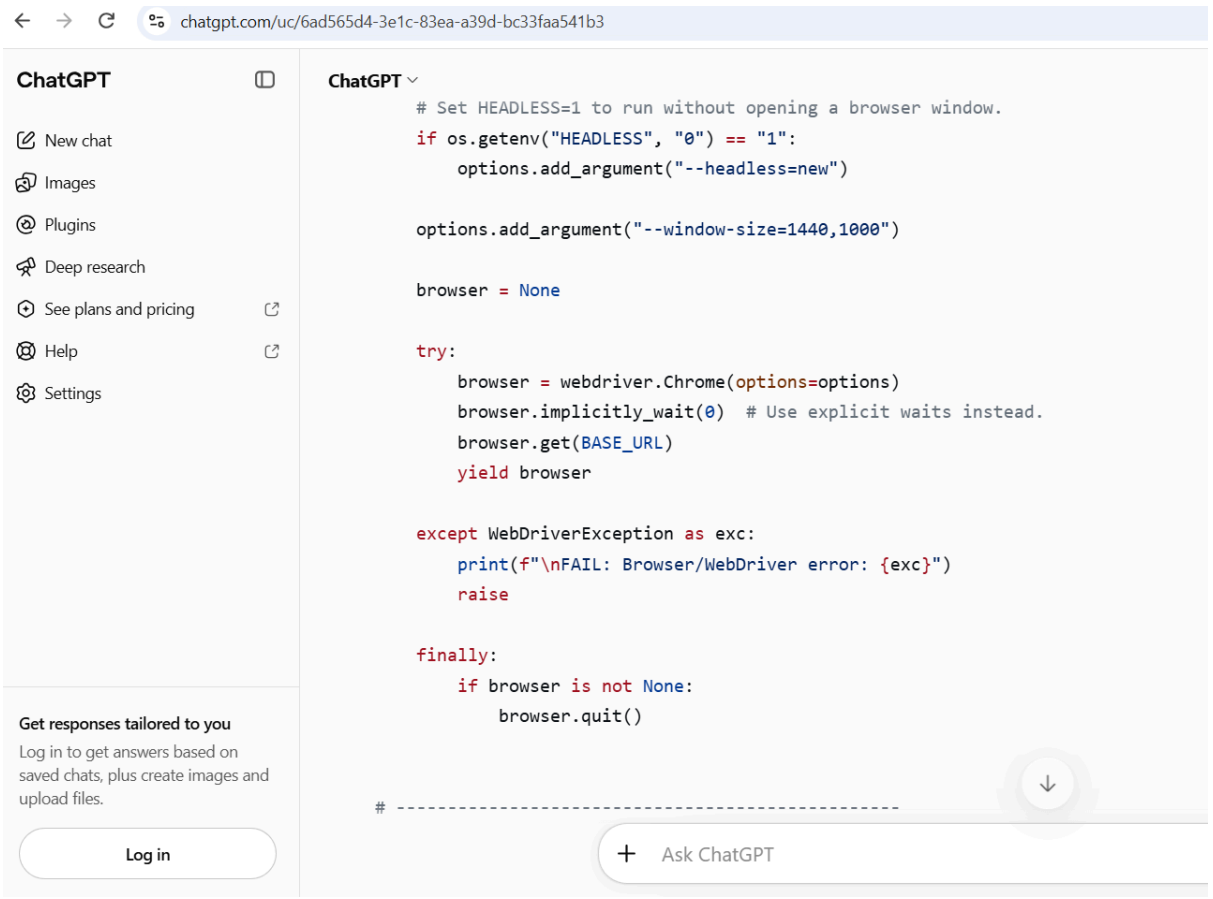
Scenario_ID:TC_04
Scenario:AI-Generated Code

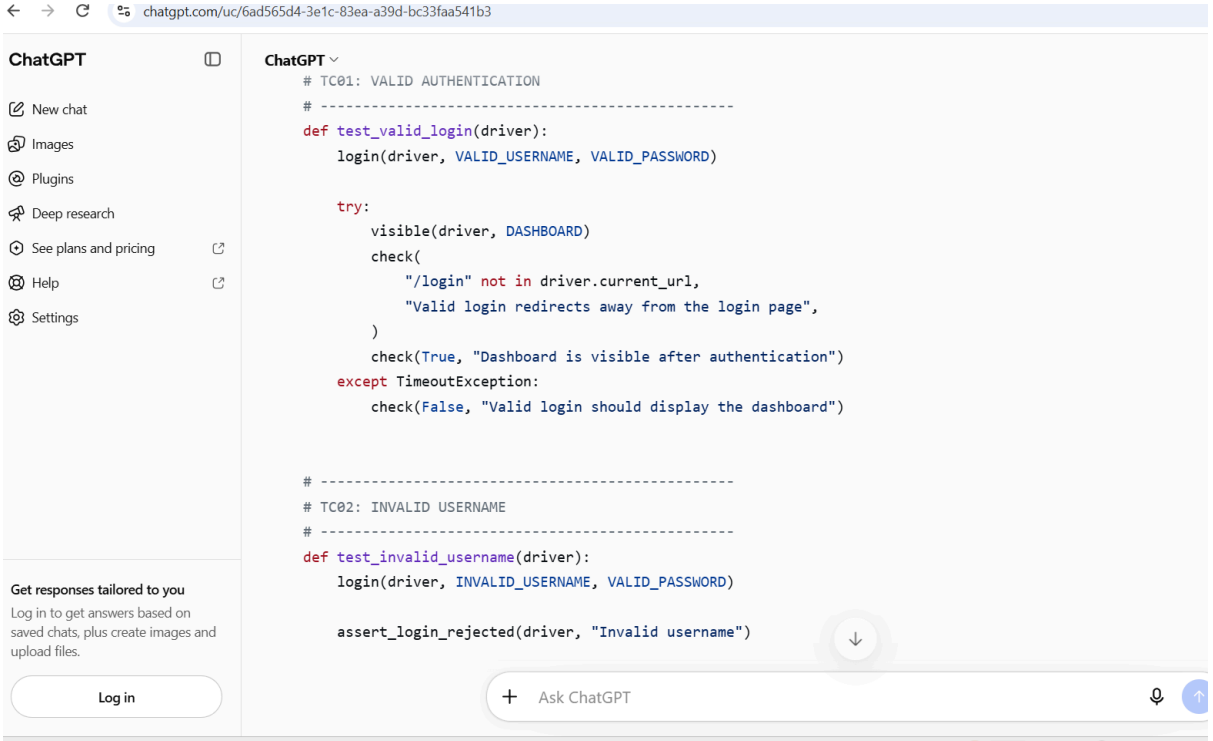

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Log in
ChatGPT
import os
import pytest
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
from selenium.common.exceptions import (
TimeoutException,
WebDriverException,
)
--
CONFIGURATION — replace with your application's values
--
BASE_URL = os.getenv(
"LOGIN_URL", "http://localhost:8000/login"
)
DASHBOARD_URL = os.getenv(
"DASHBOARD_URL", "http://localhost:8000/dashboard"
)
VALID_USERNAME = os.getenv("TEST_USERNAME", "qa_valid_user")
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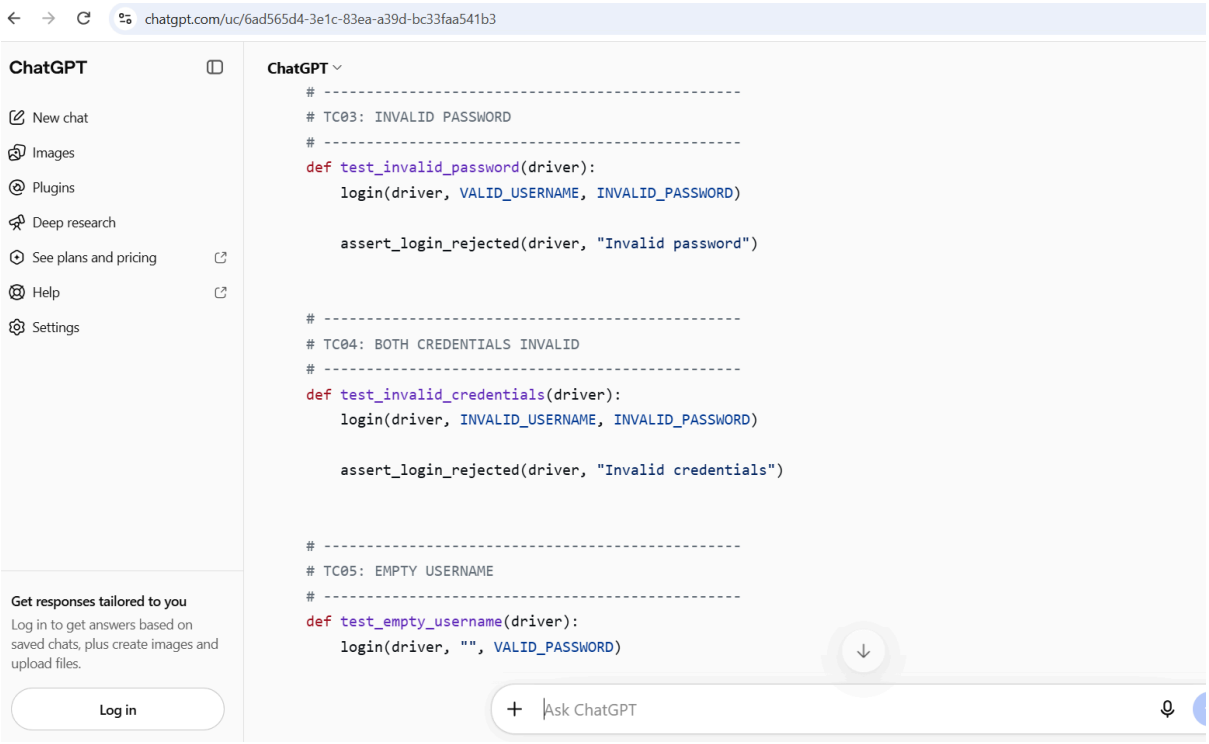

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INVALID_USERNAME = "unknown_user_999"
INVALID_PASSWORD = "WrongPass@123"
WAIT_SECONDS = 10
Example application selectors
USERNAME = (By.ID, "username")
PASSWORD = (By.ID, "password")
LOGIN_BUTTON = (By.ID, "login-button")
LOGIN_ERROR = (By.ID, "login-error")
DASHBOARD = (By.ID, "dashboard")
LOGOUT_BUTTON = (By.ID, "logout-button")
--
TEST UTILITIES
--
def check(condition, description):
"""Print PASS/FAIL and fail the pytest test if necessary."""
if condition:
print(f"\nPASS: {description}")
else:
print(f"\nFAIL: {description}")
pytest.fail(description, pytrace=False)


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def clickable(driver, locator, timeout=WAIT_SECONDS):
"""Wait until an element is clickable."""
return WebDriverWait(driver, timeout).until(
EC.element_to_be_clickable(locator)
)
def login(driver, username, password):
"""Fill and submit the login form."""
user_field = visible(driver, USERNAME)
password_field = visible(driver, PASSWORD)
user_field.clear()
user_field.send_keys(username)
password_field.clear()
password_field.send_keys(password)
clickable(driver, LOGIN_BUTTON).click()
def assert_login_rejected(driver, description):
"""Verify that login fails and an error is displayed."""
try:
Ask ChatGPT

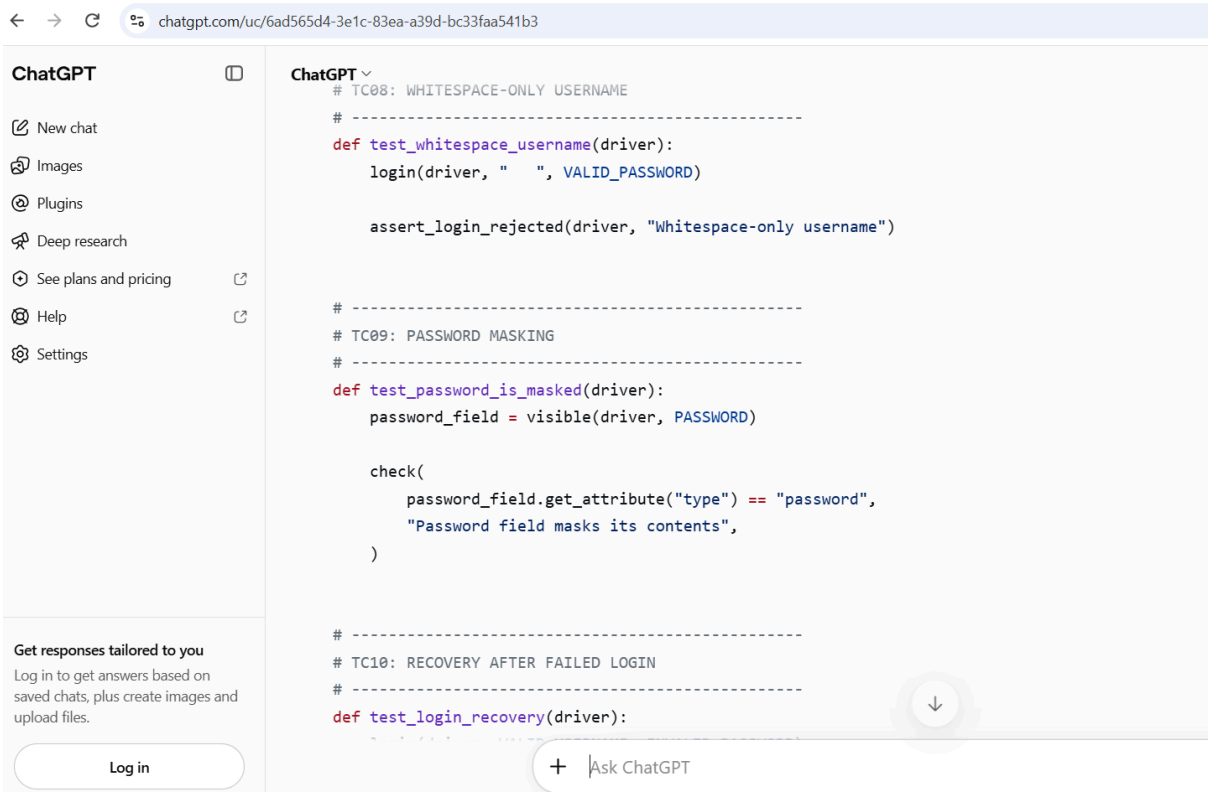

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ChatGPT
try:
error = visible(driver, LOGIN_ERROR)
check(
error.is_displayed() and bool(error.text.strip()),
f"{description}: visible error message",
)
check(
DASHBOARD not in [
(By.ID, element.get_attribute("id"))
for element in driver.find_elements(*DASHBOARD)
],
f"{description}: no dashboard element",
)
except TimeoutException:
check(False, f"{description}: error message not displayed")
--
BROWSER FIXTURE
--
@pytest.fixture
def driver():
options = webdriver.ChromeOptions()
Ask ChatGPT

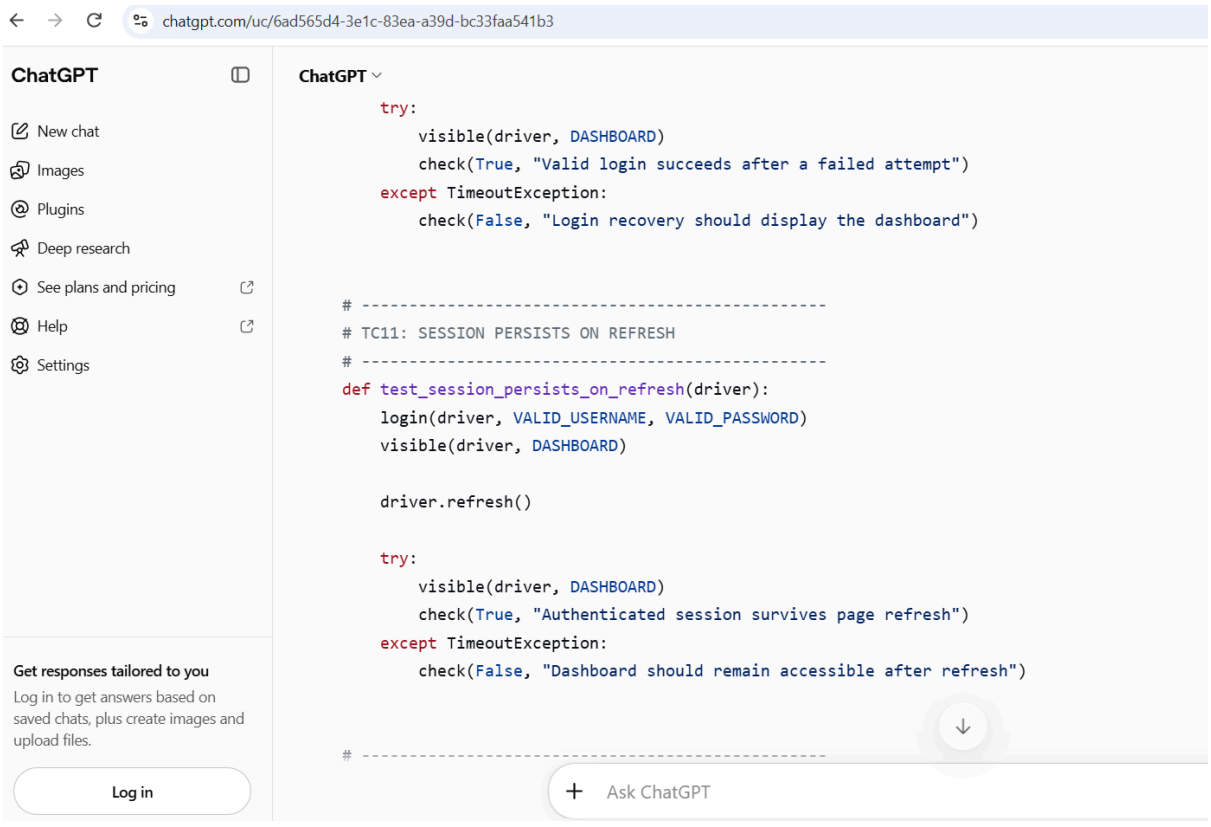

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Set HEADLESS=1 to run without opening a browser window.
if os.getenv("HEADLESS", "0") == "1":
options.add_argument("--headless=new")
options.add_argument("--window-size=1440,1000")
browser = None
try:
browser = webdriver.Chrome(options=options)
browser.implicitly_wait(0) # Use explicit waits instead.
browser.get(BASE_URL)
yield browser
except WebDriverException as exc:
print(f"\nFAIL: Browser/WebDriver error: {exc}")
raise
finally:
if browser is not None:
browser.quit()
--
Ask ChatGPT

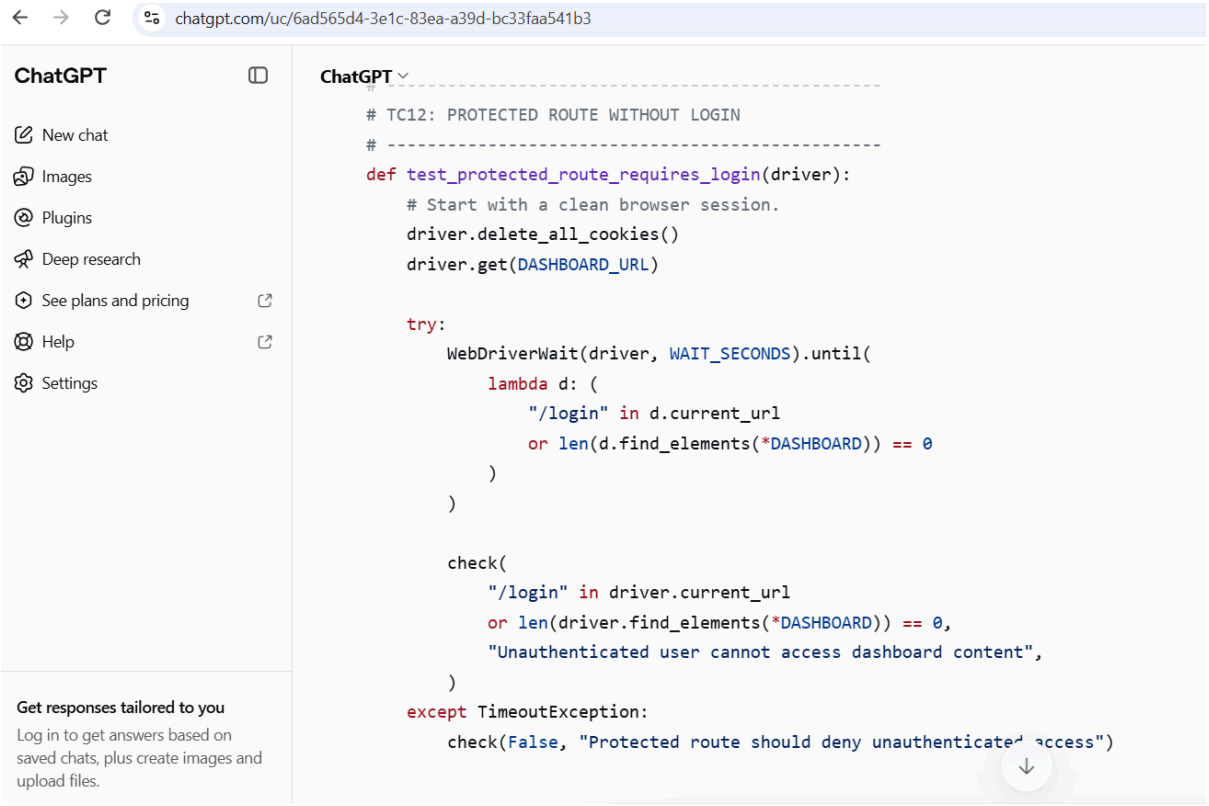

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TC01: VALID AUTHENTICATION
--
def test_valid_login(driver):
login(driver, VALID_USERNAME, VALID_PASSWORD)
try:
visible(driver, DASHBOARD)
check(
"/login" not in driver.current_url,
"Valid login redirects away from the login page",
)
check(True, "Dashboard is visible after authentication")
except TimeoutException:
check(False, "Valid login should display the dashboard")
--
TC02: INVALID USERNAME
--
def test_invalid_username(driver):
login(driver, INVALID_USERNAME, VALID_PASSWORD)
assert_login_rejected(driver, "Invalid username")
Ask ChatGPT


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--
TC03: INVALID PASSWORD
--
def test_invalid_password(driver):
login(driver, VALID_USERNAME, INVALID_PASSWORD)
assert_login_rejected(driver, "Invalid password")
--
TC04: BOTH CREDENTIALS INVALID
--
def test_invalid_credentials(driver):
login(driver, INVALID_USERNAME, INVALID_PASSWORD)
assert_login_rejected(driver, "Invalid credentials")
--
TC05: EMPTY USERNAME
--
def test_empty_username(driver):
login(driver, "", VALID_PASSWORD)
Ask ChatGPT

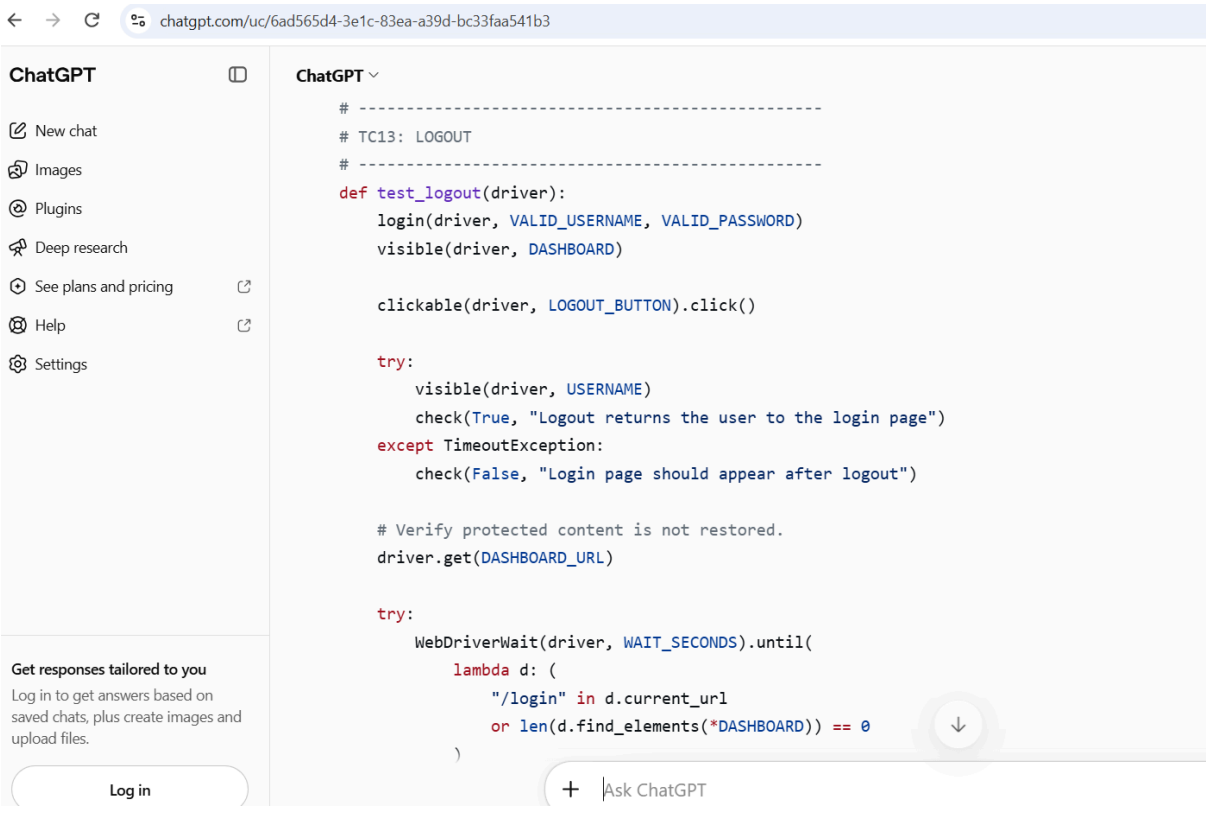

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ChatGPT
Assumes the application displays a custom error.
assert_login_rejected(driver, "Empty username")
--
TC06: EMPTY PASSWORD
--
def test_empty_password(driver):
login(driver, VALID_USERNAME, "")
assert_login_rejected(driver, "Empty password")
--
TC07: BOTH FIELDS EMPTY
--
def test_empty_login_form(driver):
login(driver, "", "")
assert_login_rejected(driver, "Empty login form")
--
Ask ChatGPT


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Log in
ChatGPT
TC08: WHITESPACE-ONLY USERNAME
--
def test_whitespace_username(driver):
login(driver, " ", VALID_PASSWORD)
assert_login_rejected(driver, "Whitespace-only username")
--
TC09: PASSWORD MASKING
--
def test_password_is_masked(driver):
password_field = visible(driver, PASSWORD)
check(
password_field.get_attribute("type") == "password",
"Password field masks its contents",
)
--
TC10: RECOVERY AFTER FAILED LOGIN
--
def test_login_recovery(driver):
Ask ChatGPT

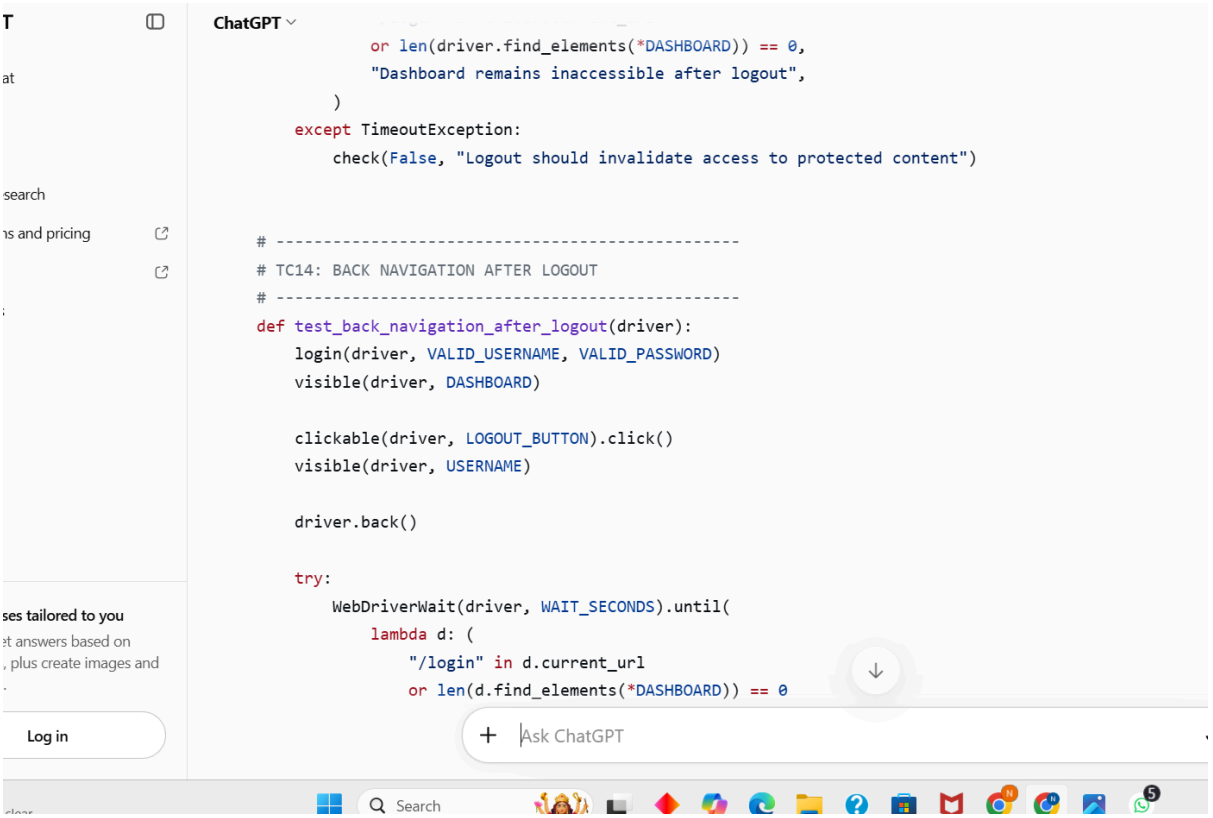

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try:
visible(driver, DASHBOARD)
check(True, "Valid login succeeds after a failed attempt")
except TimeoutException:
check(False, "Login recovery should display the dashboard")
--
TC11: SESSION PERSISTS ON REFRESH
--
def test_session_persists_on_refresh(driver):
login(driver, VALID_USERNAME, VALID_PASSWORD)
visible(driver, DASHBOARD)
driver.refresh()
try:
visible(driver, DASHBOARD)
check(True, "Authenticated session survives page refresh")
except TimeoutException:
check(False, "Dashboard should remain accessible after refresh")
--
Ask ChatGPT

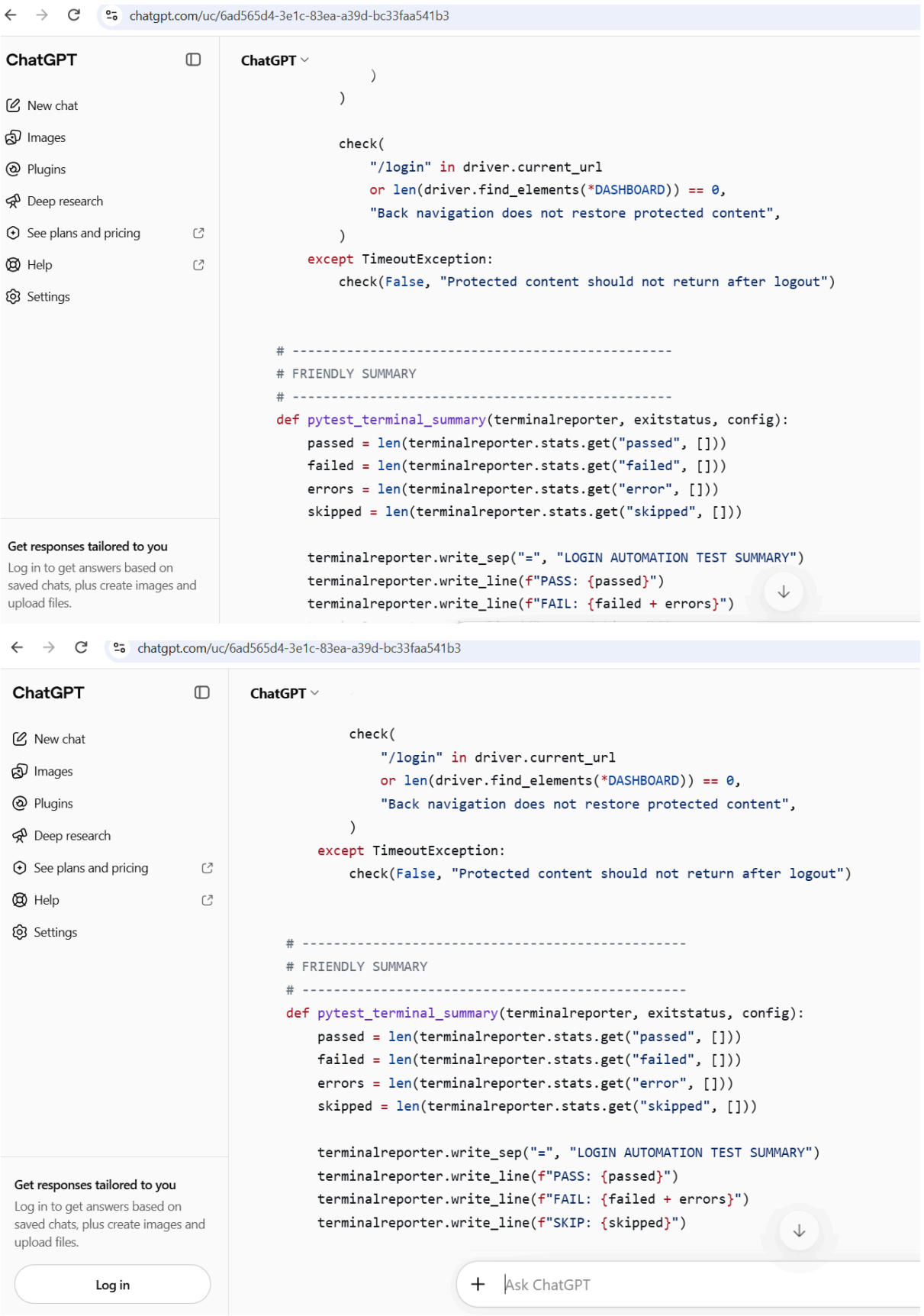

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TC12: PROTECTED ROUTE WITHOUT LOGIN
----
def test_protected_route_requires_login(driver):
Start with a clean browser session.
driver.delete_all_cookies()
driver.get(DASHBOARD_URL)
try:
WebDriverWait(driver, WAIT_SECONDS).until(
lambda d: (
"/login" in d.current_url
or len(d.find_elements(*DASHBOARD)) == 0
)
)
check(
"/login" in driver.current_url
or len(driver.find_elements(*DASHBOARD)) == 0,
"Unauthenticated user cannot access dashboard content",
)
except TimeoutException:
check(False, "Protected route should deny unauthenticated access")

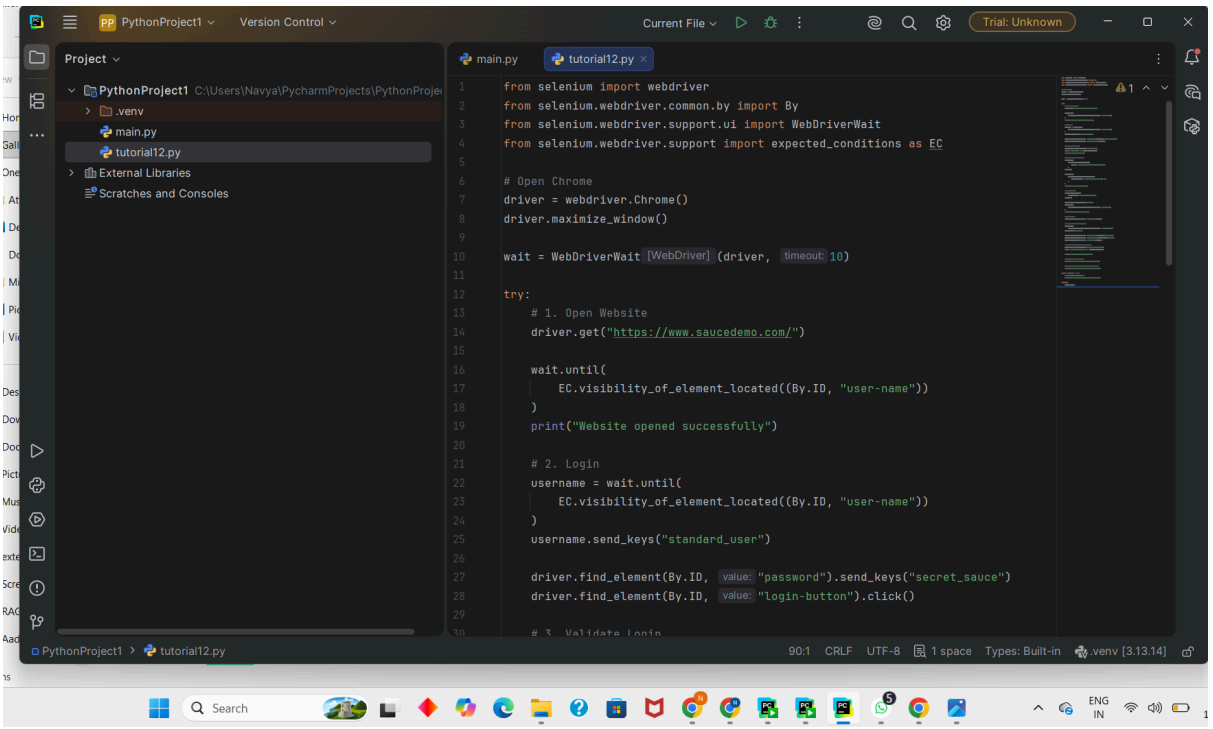

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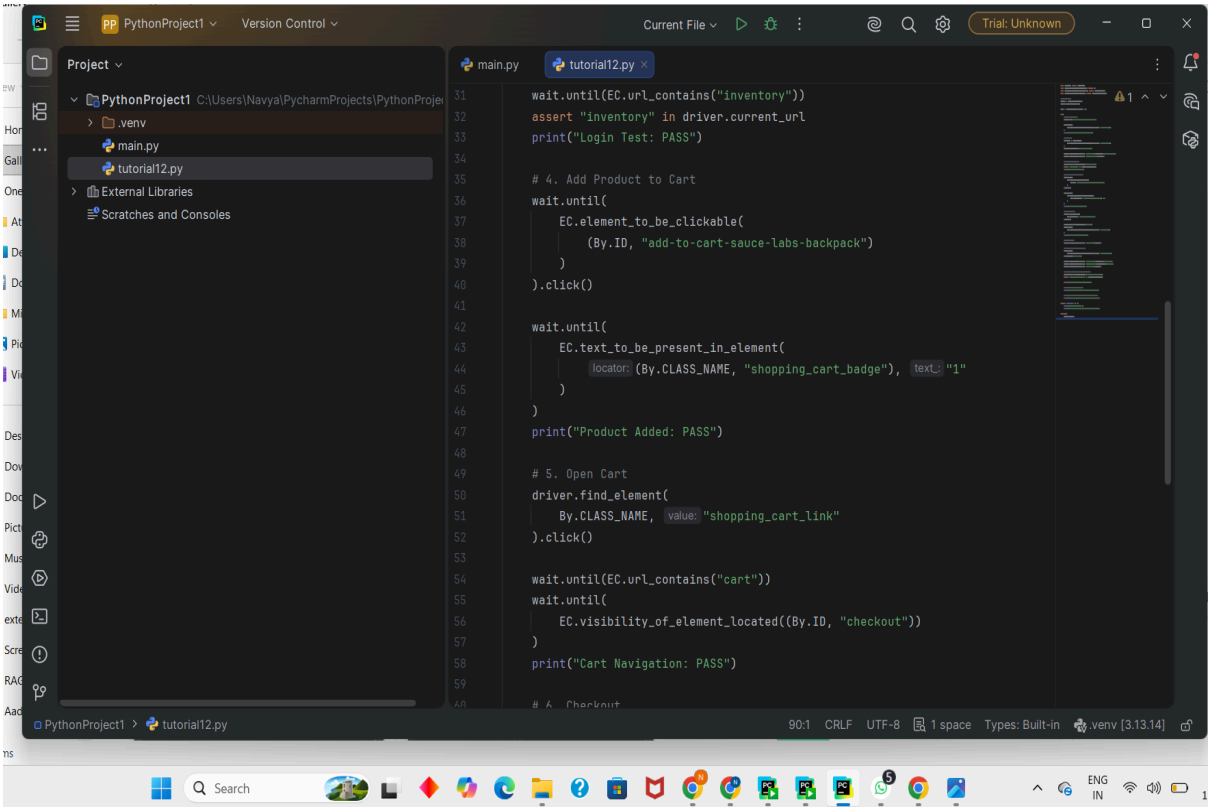
TC13: LOGOUT
----
def test_logout(driver):
login(driver, VALID_USERNAME, VALID_PASSWORD)
visible(driver, DASHBOARD)
clickable(driver, LOGOUT_BUTTON).click()
try:
visible(driver, USERNAME)
check(True, "Logout returns the user to the login page")
except TimeoutException:
check(False, "Login page should appear after logout")
Verify protected content is not restored.
driver.get(DASHBOARD_URL)
try:
WebDriverWait(driver, WAIT_SECONDS).until(
lambda d: (
"/login" in d.current_url
or len(d.find_elements(*DASHBOARD)) == 0
)
Ask ChatGPT

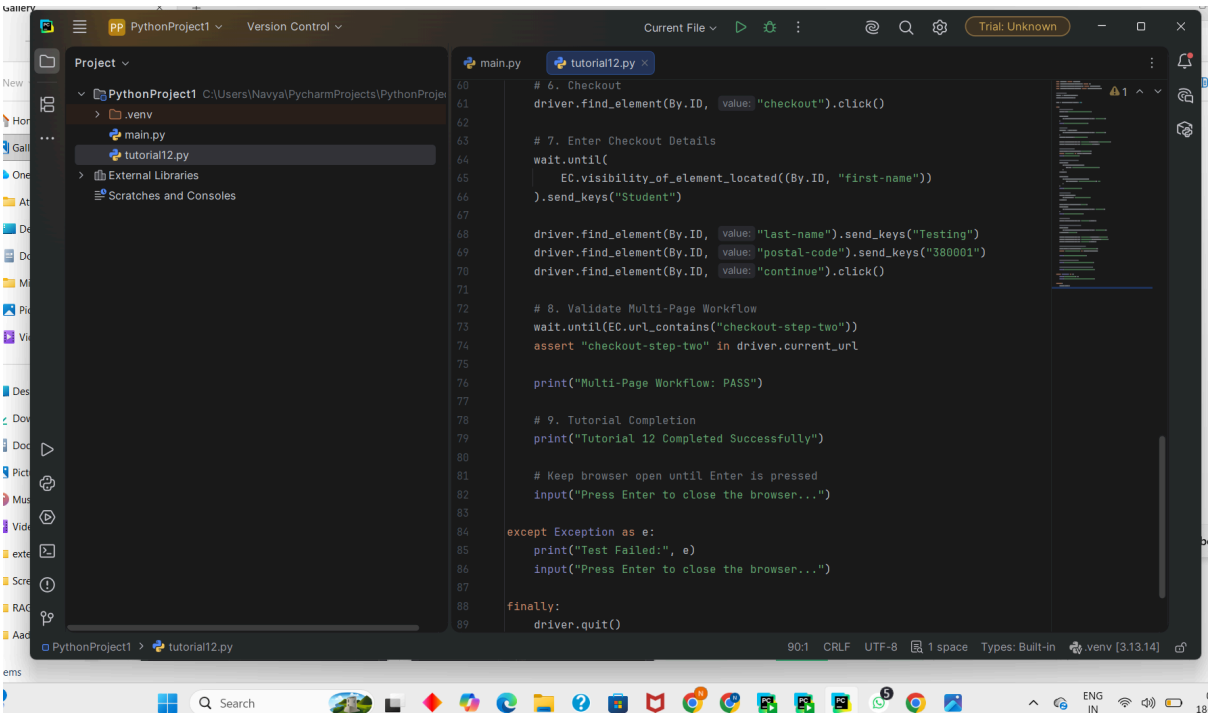

ChatGPT
or len(driver.find_elements(*DASHBOARD)) == 0,
"Dashboard remains inaccessible after logout",
)
except TimeoutException:
check(False, "Logout should invalidate access to protected content")
--
TC14: BACK NAVIGATION AFTER LOGOUT
--
def test_back_navigation_after_logout(driver):
login(driver, VALID_USERNAME, VALID_PASSWORD)
visible(driver, DASHBOARD)
clickable(driver, LOGOUT_BUTTON).click()
visible(driver, USERNAME)
driver.back()
try:
WebDriverWait(driver, WAIT_SECONDS).until(
lambda d: (
"/login" in d.current_url
or len(d.find_elements(*DASHBOARD)) == 0
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Settings
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ChatGPT
)
)
check(
"/login" in driver.current_url
or len(driver.find_elements(*DASHBOARD)) == 0,
"Back navigation does not restore protected content",
)
except TimeoutException:
check(False, "Protected content should not return after logout")
--
FRIENDLY SUMMARY
--
def pytest_terminal_summary(terminalreporter, exitstatus, config):
passed = len(terminalreporter.stats.get("passed", []))
failed = len(terminalreporter.stats.get("failed", []))
errors = len(terminalreporter.stats.get("error", []))
skipped = len(terminalreporter.stats.get("skipped", []))
terminalreporter.write_sep("=", "LOGIN AUTOMATION TEST SUMMARY")
terminalreporter.write_line(f"PASS: {passed}")
terminalreporter.write_line(f"FAIL: {failed + errors}")
chatgpt.com/uc/6ad565d4-3e1c-83ea-a39d-bc33faa541b3
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Settings
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Log in to get answers based on saved chats, plus create images and upload files.
Log in
ChatGPT
check(
"/login" in driver.current_url
or len(driver.find_elements(*DASHBOARD)) == 0,
"Back navigation does not restore protected content",
)
except TimeoutException:
check(False, "Protected content should not return after logout")
--
FRIENDLY SUMMARY
--
def pytest_terminal_summary(terminalreporter, exitstatus, config):
passed = len(terminalreporter.stats.get("passed", []))
failed = len(terminalreporter.stats.get("failed", []))
errors = len(terminalreporter.stats.get("error", []))
skipped = len(terminalreporter.stats.get("skipped", []))
terminalreporter.write_sep("=", "LOGIN AUTOMATION TEST SUMMARY")
terminalreporter.write_line(f"PASS: {passed}")
terminalreporter.write_line(f"FAIL: {failed + errors}")
terminalreporter.write_line(f"SKIP: {skipped}")
Ask ChatGPT

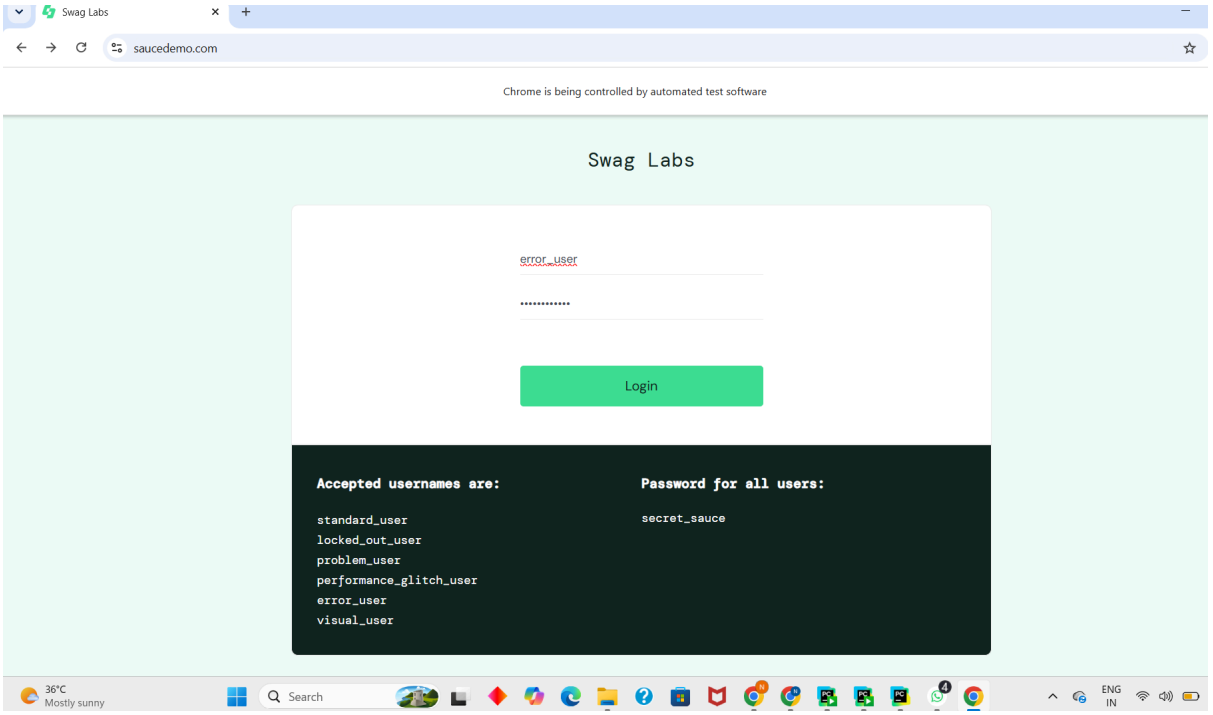
Scenario_ID:TC_05
Scenario:Correct Code in PyCharm


PythonProject1
Version Control
Current File
Trial: Unknown
Project
PythonProject1 C:\Users\Navya\PycharmProjects\PythonProje
.venv
main.py
tutorial12.py
External Libraries
Scratches and Consoles
main.py
tutorial12.py
from selenium import webdriver
from selenium.webdriver.common.by import By
from selenium.webdriver.support.ui import WebDriverWait
from selenium.webdriver.support import expected_conditions as EC
Open Chrome
driver = webdriver.Chrome()
driver.maximize_window()
wait = WebDriverWait [WebDriver] (driver, timeout: 10)
try:
1. Open Website
driver.get("https://www.saucedemo.com/")
wait.until(
EC.visibility_of_element_located((By.ID, "user-name"))
)
print("Website opened successfully")
2. Login
username = wait.until(
EC.visibility_of_element_located((By.ID, "user-name"))
)
username.send_keys("standard_user")
driver.find_element(By.ID, value: "password").send_keys("secret_sauce")
driver.find_element(By.ID, value: "login-button").click()
PythonProject1 > tutorial12.py
90:1 CRLF UTF-8 1 space Types: Built-in .venv [3.13.14]
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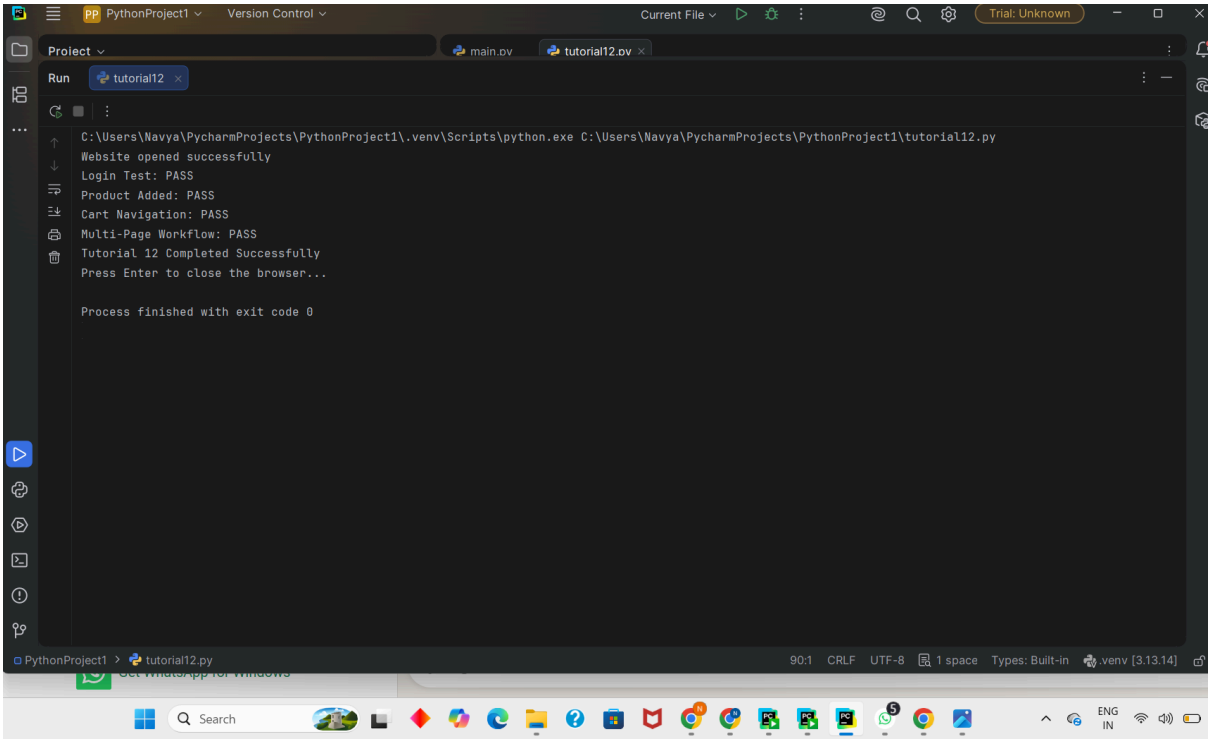

PythonProject1
Version Control
Current File
Trial: Unknown
Project
PythonProject1 C:\Users\Navya\PycharmProjects\PythonProje
.venv
main.py
tutorial12.py
External Libraries
Scratches and Consoles
main.py
tutorial12.py
wait.until(EC.url_contains("inventory"))
assert "inventory" in driver.current_url
print("Login Test: PASS")
4. Add Product to Cart
wait.until(
EC.element_to_be_clickable(
(By.ID, "add-to-cart-sauce-labs-backpack")
)
).click()
wait.until(
EC.text_to_be_present_in_element(
locator: (By.CLASS_NAME, "shopping_cart_badge"), text_: "1"
)
)
print("Product Added: PASS")
5. Open Cart
driver.find_element(
By.CLASS_NAME, value: "shopping_cart_link"
).click()
wait.until(EC.url_contains("cart"))
wait.until(
EC.visibility_of_element_located((By.ID, "checkout"))
)
print("Cart Navigation: PASS")
PythonProject1 > tutorial12.py
90:1 CRLF UTF-8 1 space Types: Built-in .venv [3.13.14]
Search
ENG
IN



Scenario_ID:TC_06
Scenario:Browser Output



Scenario_ID:TC_07
Scenario:Console Output



Conclusion:

AI speeds up test design and script writing, but its output must be reviewed and corrected by a tester before it is trusted.

Summary:

A generative AI tool was given the login requirement and produced test cases, test data, defect-prone areas and Selenium code. The code was reviewed, corrected and run for the valid-login case; the regression cases were then re-checked.